

- All problems are worth an equal weight, even if some may be harder than others.
 - Same rules as in the past, except for **choose 5 out of the 6 problems below to submit**.
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1. For both differential equations below,

$$\frac{d^2y}{dt^2} + 4y = \cos\left(\frac{9t}{4}\right), \quad \frac{d^2y}{dt^2} + 5y = 3\cos(2t).$$

- determine the frequency of the beats.
- determine the frequency of the rapid oscillations, and
- use the information from parts (a) and (b) to give a rough sketch of a typical solution.

2. Consider the system

$$\begin{cases} \frac{dx}{dt} = -2x + y \\ \frac{dy}{dt} = -y + x^2 \end{cases}$$

- Find the linearized system for the equilibrium point $(0, 0)$.
- Classify $(0, 0)$ (as either a source, sink, center, ...)
- Sketch the phase portrait for the linearized system near $(0, 0)$.
- Repeat parts (a)–(c) for the equilibrium point at $(2, 4)$.

3. Suppose two species X and Y are to be introduced to an island. It is known that the two species compete, but the precise nature of their interaction is unknown. We assume that the populations $x(t)$ and $y(t)$, respectively, are modeled by a system

$$\begin{cases} \frac{dx}{dt} = f(x, y) \\ \frac{dy}{dt} = g(x, y). \end{cases}$$

- Suppose $f(0, 0) = g(0, 0) = 0$; that is, $(0, 0)$ is an equilibrium point. What does this say about the ability of X and Y to migrate to the island?
- Suppose that a small population consisting solely of one species reproduces rapidly. What can you conclude about the values of $\partial f/\partial x$ and $\partial g/\partial y$ at $(0, 0)$?
- Since X and Y compete for resources, the presence of either of the species will decrease the rate of growth of the population of the other. What does this say about $\partial f/\partial y$ and $\partial g/\partial x$ at $(0, 0)$?
- Using the assumptions from parts (a)–(c), what type(s) of equilibrium point could $(0, 0)$ possibly be?

4. Consider the system

$$\begin{cases} \frac{dx}{dt} = y \\ \frac{dy}{dt} = x - x^2 \end{cases}$$

- a) Sketch the nullclines and find the direction of the vector field along the nullclines.
- b) Find the linearized system near the equilibrium points $(0, 0)$ and $(1, 0)$. Try a sketch of the phase portrait with what you know.

5. For the system

$$\begin{cases} \frac{dx}{dt} = y \\ \frac{dy}{dt} = x^3 - x \end{cases}$$

- a) show that the system is Hamiltonian with Hamiltonian function

$$H(x, y) = \frac{y^2}{2} + \frac{x^2}{2} - \frac{x^4}{4}.$$

- b) sketch the level sets of H

Remark: you may use a graphing tool for this, for instance, MatLab assignment 4 can be adapted here

- c) sketch the phase portrait of the system.

6. Consider the system

$$\begin{cases} \frac{dx}{dt} = -x^3 \\ \frac{dy}{dt} = -y^3 \end{cases}$$

- a) Verify that $L(x, y) = (x^2 + y^2)/2$ is a Lyapunov function for the system.
- b) Sketch the level sets of L .
- c) What can you conclude about the phase portrait of the system from the information given in parts (a)–(b)?